Huawei Access Node YEL-ACC-67 Hardware/Environmental Fault Resolution SOP

Node: YEL-ACC-67 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_7201

Description

The YEL-ACC-67 access node, utilizing Huawei's Access equipment, is susceptible to hardware/environmental faults due to the inherent design of its ASIC memory parity error detection and correction mechanism. The Access equipment's line card crashes and power supply redundancy loss are potential failure domains within this role. The alarm code ALARM_7201, with a severity level of warning, triggers this SOP. The line card crash can be attributed to a combination of factors, including but not limited to, high temperature, high humidity, or power quality issues. These conditions can cause the line card's ASIC (Application-Specific Integrated Circuit) to fail, leading to parity errors and subsequent crashes. Additionally, the loss of power supply redundancy can also trigger the ALARM_7201, as the Huawei Access equipment relies on dual power supplies to ensure continuous operation. This redundancy is designed to prevent single points of failure and ensure high availability. However, if one power supply fails, the system will trigger the ALARM_7201, indicating a warning condition.

Pre-requisite Checks

- id: 1, check: Verify the site location (Yelahanka Sub-station) and access node ID (YEL-ACC-67) to ensure correct equipment identification.

- id: 2, check: Check the redundant link status to confirm the loss of redundancy is not a result of intentional configuration changes.

- id: 3, check: Ensure remote hands are dispatched to the Yelahanka Sub-station to assist with physical checks and repairs.

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to high temperature and high humidity conditions	High
Line card crashes due to power quality issues	Medium
Loss of power supply redundancy	Low

Diagnostic Commands & Expected Outputs

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Command: display current-temperature, Expected Output: Current temperature of the line card: 45.6°C, Description: This command retrieves the current temperature of the line card, which can help identify potential overheating issues.
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Command: display alarm-information ALARM_7201, Expected Output: ALARM_7201: Warning, Line card crashed due to parity error, Description: This command retrieves the alarm information for the ALARM_7201, which can help confirm the cause of the alarm.

Command: display power-supply information, Expected Output: Power supply 1: Normal, Power supply 2: Faulty, Description: This command retrieves the power supply information, which can help identify potential power quality issues.
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Resolution Steps

Step 1: Step: Replace the faulty line card, Command: replace line-card Line-Card-1, Description: This command initiates the replacement of the faulty line card with a new one.

Step 2: Step: Replace the faulty power supply, Command: replace power-supply Power-Supply-2, Description: This command initiates the replacement of the faulty power supply with a new one.

Step 3: Step: Verify the system is operational, Command: verify system operational, Description: This command verifies that the system is operational and the alarm is cleared.

Post-Check Verification

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Command: display current-temperature, Expected Output: Current temperature of the line card: 35.6°C, Description: This command verifies that the line card temperature is within normal operating range.

Command: display alarm-information ALARM_7201, Expected Output: ALARM_7201: Cleared, Line card replaced, Description: This command verifies that the ALARM 7201 is cleared and the line card is replaced.

Command: display power-supply information, Expected Output: Power supply 1: Normal, Power supply 2: Normal, Description: This command verifies that both power supplies are operational.
```

Roll-back Procedures

- Step: Restore the original line card, Command: restore line-card Line-Card-1, Description: This command restores the original line card if the replacement fails.

- Step: Restore the original power supply, Command: restore power-supply Power-Supply-2, Description: This command restores the original power supply if the replacement fails.